



Modelling Financial Markets

Chapter 0: Financial Markets in 2026



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Acronyms used in this material (1/2)

- ▣ AMM: Automated Market Maker
- ▣ AP: Authorised Participant
- ▣ ASE: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-FI: BET Financial Investment companies index
- ▣ BET-TR: BET Total Return (index)
- ▣ BIS: Bank for International Settlements
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CBOE: Chicago Board Options Exchange
- ▣ CFTC: Commodity Futures Trading Commission
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CRIX: CRypto IndeX
- ▣ EBA: European Banking Authority
- ▣ EMH: Efficient Market Hypothesis
- ▣ ESMA: European Securities and Markets Authority
- ▣ ETF: Exchange-Traded Fund
- ▣ ETP: Exchange-Traded Product
- ▣ EU: European Union
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FX: Foreign eXchange
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GENIUS: Guiding and Establishing National Innovation for U.S. Stablecoins (Act)
- ▣ HFT: High-Frequency Trading

Acronyms used in this material (2/2)

- IPO: Initial Public Offering
- NAV: Net Asset Value
- NYSE: New York Stock Exchange
- OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- OTC: Over-The-Counter
- ROL: Romanian old leu (ISO 4217 code, until 2005)
- RWA: Real-World Assets
- SEC: Securities and Exchange Commission
- SPDR: Standard & Poor's Depositary Receipts
- UDR: Uzinele și Domeniile Reșița (the Reșița Works and Domains)
- US: United States
- UTC: Coordinated Universal Time (Temps Universel Coordonné)
- VIX: CBOE Volatility Index

Outline

Course Organisation

Key Measures Used in This Chapter

The Financial System

Market Structure

The 2000–2026 Landscape

Passive Investing and Concentration

Digital Assets and Tokenisation

The Romanian Market

Data and Tools

Conclusions

References

About the Course

Programme

- Master's programme **Applied Statistics and Data Science**
- Faculty of Cybernetics, Statistics and Economic Informatics, Bucharest University of Economic Studies (ASE)
- 2 hours lecture + 2 hours seminar per week, academic year 2026/2027

Team

- Lecture: Prof. Daniel Traian Pele, danpele@ase.ro
- Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro
- Office hours: by appointment

What You Will Be Able to Do

- Describe how modern markets work
 - ▶ from exchanges and ETFs to tokenised assets and cryptocurrencies
- Build, estimate and test statistical models on real data
 - ▶ returns, volatility, risk, dependence and bubbles
- Evaluate models honestly
 - ▶ out-of-sample tests, backtests and proper scoring of forecasts
- Produce reproducible research in Python and GitHub
 - ▶ every chart documented as a Quantlet: code + description of inputs and outputs

Course Map: 20 Chapters

- ▣ 0 Financial Markets in 2026
- ▣ 1 Stylized Facts and Returns
- ▣ 2 Market Efficiency: From EMH to Adaptive Markets
- ▣ 3 Factor Models and Asset Pricing
- ▣ 4 Portfolio Optimization
- ▣ 5 Conditional Volatility: GARCH Models
- ▣ 6 Multivariate Volatility and Dependence
- ▣ 7 Value-at-Risk and Expected Shortfall
- ▣ 8 Backtesting and Evaluating Risk Forecasts
- ▣ 9 Realized Volatility
- ▣ 10 Market Microstructure
- ▣ 11 Continuous-Time Models
- ▣ 12 Options and the Volatility Surface
- ▣ 13 Machine Learning in Finance
- ▣ 14 Deep Learning and Time-Series Foundation Models
- ▣ 15 LLMs and Sentiment Analysis
- ▣ 16 Digital Assets and DeFi
- ▣ 17 Bubbles and Crashes
- ▣ 18 Systemic Risk and Networks
- ▣ 19 Review and Project Presentations

Assessment

Grading

- ▣ Team project (GitHub + presentation): **70%**
- ▣ Quizzes and seminar activity: **20%**
- ▣ Attendance: **10%**
- ▣ No written exam

Quizzes

- ▣ One quiz per chapter, on the course website
- ▣ 20 questions drawn at random from a larger bank
- ▣ Each answer is explained immediately

The Team Project

- Teams of 3–4 students, one research question on real market data
- Public GitHub repository
 - ▶ code, a description of the data, results and a README file explaining how to run everything
- Every figure published as a Quantlet
 - ▶ one folder per result: code + `Metainfo.txt` (name, description, keywords, inputs, outputs) + chart
- Weekly commits: the history of the repository shows how the work progressed
- Final presentation in Chapter 19

Tools

Software

- ▣ Python with pandas, numpy, scipy, statsmodels, scikit-learn, matplotlib
- ▣ Jupyter notebooks, runnable in Google Colab without installation
- ▣ Git and GitHub for code, data and version history

Platforms

- ▣ **Quantlet**: a repository of documented code, one folder per chart
- ▣ **Quantinar**: video courses for advanced topics
- ▣ Course repository: github.com/danpele/MFM

Reproducibility Is Part of the Result

- Rule of the course: a result exists only if someone else can reproduce it from your repository
- In practice
 - ▶ the data are loaded by code, never copied manually
 - ▶ every number in a slide or report comes from a script
 - ▶ random numbers use a fixed seed, so reruns give the same result
- All charts of this lecture follow the rule
 - ▶ each chart has a Quantlet link at the bottom right of its slide

Core Reading

- ▣ Franke, Härdle & Hafner (2019), *Statistics of Financial Markets*, 5th ed.: the main statistical reference
- ▣ Campbell, Lo & MacKinlay (1997), *The Econometrics of Financial Markets*
- ▣ Brandimarte (2018), *An Introduction to Financial Markets: A Quantitative Approach*
- ▣ Härdle & Simar (2019), *Applied Multivariate Statistical Analysis*
- ▣ Härdle, Chen & Overbeck (2017) (Eds.), *Applied Quantitative Finance*, 3rd ed.
- ▣ Härdle, Harvey & Reule (2020), "Understanding cryptocurrencies", *Journal of Financial Econometrics*
 - ▶ Further papers are cited in each chapter; every citation is a clickable link

Prices, Indices and Total Return

- **Price** P_t : the closing price of an asset on day t
- **Index**: a weighted average of many prices, e.g. the S&P 500 (500 large US companies)
- **Price index**: follows prices only; dividends paid by the companies are ignored
- **Total-return index**: assumes that every dividend is reinvested in the index
 - ▶ Example: BET is a price index of the Bucharest Stock Exchange, BET-TR is its total-return version
- **Adjusted close**: a stock or fund price corrected for dividends and splits, so that it behaves like a total-return series

Simple and Log Returns

Definition 1 (Returns)

□ Simple return: $R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$

□ Log return: $r_t = \ln P_t - \ln P_{t-1} = \ln(1 + R_t)$

- For small moves the two are almost equal: $r_t \approx R_t$ when $|R_t|$ is a few percent
- Log returns add over time: $r_{t-1} + r_t = \ln(P_t/P_{t-2})$
- Chapter 1 studies returns in detail; here they are the input of the measures below

Compound Annual Growth Rate (CAGR)

Definition 2 (CAGR)

- The constant yearly growth rate that turns the start value into the end value:

$$\text{CAGR} = \left(\frac{P_{\text{end}}}{P_{\text{start}}} \right)^{1/T} - 1, \quad T = \frac{\text{calendar days between the two dates}}{365.25}$$

- Also called the **annualised return**; it ignores the path, only the start and the end matter
- “x3.7” in a chart means $P_{\text{end}}/P_{\text{start}} = 3.7$: the value of 1 USD invested at the start

Worked Example: CAGR of the S&P 500, 2015–2026

- Start: 2 January 2015, $P_{\text{start}} = 2\,058.20$
- End: 18 September 2026, $P_{\text{end}} = 7\,650.50$
- Growth multiple: $7\,650.50 / 2\,058.20 = 3.717$ (“x3.7”)
- Length: $T = 4\,277 / 365.25 = 11.71$ years
- CAGR: $3.717^{1/11.71} - 1 = 0.119$, i.e. **11.9% per year**
 - ▶ Check: $1.119^{11.71} \approx 3.72$
 - ▶ Price index only: with dividends reinvested (SPY) the CAGR is 13.7%, i.e. 1.8 pp higher

 MFM_ch0_cross_asset

Annualised Volatility

Definition 3 (Volatility)

- Daily volatility: the standard deviation of daily log returns, $\hat{\sigma}_d = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (r_t - \bar{r})^2}$
- Annualised volatility: $\hat{\sigma}_a = \hat{\sigma}_d \sqrt{N}$, with N = trading days per year
- $N = 252$ for stocks, bonds and currencies (working days), $N = 365$ for crypto (trades every day)
- Example, S&P 500 2015–2026: $\hat{\sigma}_d = 1.115\%$, so $\hat{\sigma}_a = 1.115\% \times \sqrt{252} = 17.7\%$
- The square root follows from summing independent daily returns: variances add up

Drawdown and Maximum Drawdown

Definition 4 (Drawdown)

- ▣ Running peak: $M_t = \max_{s \leq t} P_s$
 - ▣ Drawdown: $DD_t = P_t/M_t - 1 \leq 0$, the loss from the previous peak
 - ▣ Maximum drawdown: $MDD = \min_t DD_t$, the worst peak-to-trough loss
-
- ▣ Example, S&P 500 in the global financial crisis
 - ▶ peak 1 565.15 (9 Oct 2007), trough 676.53 (9 Mar 2009): $676.53/1\,565.15 - 1 = -56.8\%$
 - ▶ the index regained its 2007 peak only on 28 Mar 2013, five and a half years later

Correlation and Rolling Correlation

Definition 5 (Correlation)

- Pearson correlation of two return series: $\rho_{XY} = \frac{\text{Cov}(r^X, r^Y)}{\sigma_X \sigma_Y} \in [-1, 1]$
 - Rolling correlation: the same formula on a moving window, e.g. the last 252 trading days (one year)
-
- $\rho > 0$: the assets tend to move together; $\rho < 0$: one tends to rise when the other falls
 - Diversification works best with low or negative correlation
 - Assets with different calendars (stocks vs. crypto) are compared only on common trading days

Yields, Spreads, Percentage Points and Basis Points

- **Yield:** the annual return of a bond bought today and held to maturity, in % per year
- **Yield-curve spread:** $s_t = y_t^{10y} - y_t^{2y}$, the 10-year minus the 2-year government yield
- **Inversion:** $s_t < 0$, short-term rates above long-term rates, historically often followed by recessions
- Differences of rates are measured in
 - ▶ **percentage points (pp):** $4.94\% - 4.67\% = 0.27$ pp
 - ▶ **basis points (bp):** $1 \text{ bp} = 0.01 \text{ pp}$, so $0.27 \text{ pp} = 27 \text{ bp}$
- Example: US Treasuries on 17 Sep 2026 (next section)

Recap: Key Measures

Measure	Formula	Reads as
Log return	$r_t = \ln(P_t/P_{t-1})$	daily change, additive over time
CAGR	$(P_{\text{end}}/P_{\text{start}})^{1/T} - 1$	constant yearly growth
Annualised volatility	$\hat{\sigma}_d \sqrt{N}$, $N = 252$ or 365	typical yearly fluctuation
Drawdown	$P_t / \max_{s \leq t} P_s - 1$	loss from the last peak
Correlation	$\text{Cov}(r^X, r^Y) / (\sigma_X \sigma_Y)$	co-movement, from -1 to 1
Spread	$y^{10y} - y^{2y}$ (pp or bp)	slope of the yield curve

- All numbers in the rest of the chapter use these definitions

What Financial Markets Are

Definition 6 (Financial market)

- A mechanism, physical or electronic, where buyers and sellers exchange claims on future cash flows at prices formed by supply and demand
- A **claim** can be a share of a company, a bond, a currency, a contract or a token
- The **price** is the only number that everyone sees: it summarises the views and needs of all participants
- Statistics of financial markets studies how these prices move

What Financial Markets Do

- ▣ **Capital allocation:** savings flow to firms and governments
- ▣ **Risk sharing:** risk is transferred to those willing to bear it
- ▣ **Liquidity:** assets can be converted to cash quickly
- ▣ **Price discovery:** prices aggregate dispersed information
- ▣ **Payments and settlement:** transfer of ownership and cash
- ▣ **Monitoring:** prices discipline managers and governments

Asset Classes

Class	Examples	Cash flow	Main risks
Equities	shares, equity ETFs	dividends, residual claim	market, idiosyncratic
Fixed income	government and corporate bonds	coupons, principal	interest rate, credit
Money market	T-bills, repo	short-term interest	liquidity
FX	EUR/USD, EUR/RON	interest differential	currency, policy
Commodities	gold, oil futures	none (roll yield)	supply, storage
Derivatives	futures, options, swaps	contingent payoffs	leverage, counterparty
Digital assets	Bitcoin, Ether, stablecoins	none / protocol-specific	volatility, regulation

- The course models all of them with the same statistical toolbox: returns, volatility, dependence and tails

Asset Classes: Terms Used in the Table

- ▣ **ETF**: exchange-traded fund, a fund whose shares trade on an exchange like a stock
- ▣ **T-bills**: Treasury bills, government debt with maturity below one year
- ▣ **Repo**: repurchase agreement, a short-term loan backed by securities
- ▣ **FX**: foreign exchange, the currency market; EUR/RON is the price of one euro in lei
- ▣ **Futures**: a contract to buy or sell later at a price fixed today; **roll yield** is the gain or loss from replacing an expiring contract with the next one
- ▣ **Options**: the right, not the obligation, to buy or sell at a fixed price; **swaps**: exchanges of cash-flow streams
- ▣ **Idiosyncratic risk**: specific to one company; **counterparty risk**: the other side of a contract may not pay

Market Participants: the Buy Side

- The **buy side** invests money and buys market services
- Retail investors: individuals trading through a broker
- Institutional investors: pension funds, insurers, sovereign wealth funds
- Asset managers: mutual funds and ETFs, passive or active
 - ▶ **passive**: replicate an index; **active**: try to beat it
- Hedge funds and proprietary trading firms: trade with their own capital, often with leverage

Market Participants: Sell Side and Infrastructure

- The **sell side** provides trading services
 - ▶ **brokers** execute orders for clients; **dealers** trade from their own inventory
 - ▶ **market makers** continuously quote a buy and a sell price and earn the difference
- **Infrastructure**
 - ▶ exchanges and **multilateral trading facilities (MTFs)**: platforms where many buyers and sellers meet
 - ▶ **central counterparties (CCPs)**: become the buyer to every seller and the seller to every buyer
 - ▶ **custodians**: keep securities safe; regulators and central banks set the rules
- Participants have different horizons and information, which is why prices move in ways that simple models miss

Where Trading Happens

Exchanges

- Central limit order book
- Quotes and trades are public
- Examples: NYSE, Nasdaq, BVB

OTC markets

- Bilateral trades with dealers
- Dominant for bonds, FX, swaps
- Prices less visible

Dark pools

- No quotes shown before the trade
- Large orders hide their size
- Prices referenced to public markets

- **OTC**: over the counter, trades negotiated directly between two parties, without an exchange
- NYSE: New York Stock Exchange; BVB: Bursa de Valori București (Bucharest Stock Exchange)

Crypto Trading Venues

- **Centralised exchanges:** companies that run an order book, like a stock exchange, but open 24 hours a day, 7 days a week
- **Decentralised exchanges:** programs on a blockchain (smart contracts) that trade without an operator
 - ▶ most use an **automated market maker (AMM)**: a pool of two tokens whose price is set by a formula (Chapter 16)
- The same coin trades on hundreds of venues at slightly different prices
- **Makarov & Schoar (2020)**: large and persistent price differences across exchanges and countries
 - ▶ fragmentation creates arbitrage opportunities that capital controls and costs keep open

The Limit Order Book



- **Limit order book:** the list of all waiting buy and sell orders, sorted by price
- **Bid:** an offer to buy; the best bid is the highest one (99.9 here)
- **Ask:** an offer to sell; the best ask is the lowest one (100.01 here)
- **Bid-ask spread:** best ask minus best bid, the price of trading immediately
- **Depth:** the quantity waiting near the best prices
 - ▶ Illustration only; Chapter 10 models the book with real data

Order Types

- **Limit order:** buy or sell at a given price or better
 - ▶ waits in the book and **provides** liquidity to others
- **Market order:** buy or sell now, at the best available prices
 - ▶ executes immediately and **consumes** liquidity
- **Stop order:** becomes a market order once a trigger price is reached
 - ▶ used to limit losses; many stops at the same level can accelerate a fall
- **Liquidity:** the ability to trade a large quantity quickly without moving the price much

Worked Example: a Market Order Walks the Book

- Book of the previous slide: asks of 2 units at 100.01 and 4 units at 100.02
- A trader sends a **market buy order for 5 units**
 - ▶ 2 units are bought at 100.01, the whole first level
 - ▶ the remaining 3 units at 100.02, the next level
- Average price: $(2 \times 100.01 + 3 \times 100.02)/5 = 100.016$
- After the trade the best ask is 100.02 and the spread widens from 0.11 to 0.12
 - ▶ large orders move the price against the trader: this is **market impact** (Chapter 10)

Auctions versus Continuous Trading

Call auctions

- ▣ Orders are collected, then matched at one price that maximises traded volume
- ▣ Used at the open and close of most exchanges
- ▣ Concentrate liquidity and reduce noise in reference prices

Continuous trading

- ▣ Every incoming order is matched immediately if possible
- ▣ Priority: best price first, then the earliest order
- ▣ Speed matters: being first in the queue is rewarded

Why Market Structure Matters for Statistics

- For exchange-traded assets, the “daily close” used in this course is the price of the closing auction
 - ▶ exceptions: Bitcoin trades 24/7 with no auction, so its daily price is a snapshot at 00:00 UTC; EUR/RON is the BNR fixing, published around 13:00 Bucharest time (Chapter 1)
- **High-frequency trading (HFT)**: computer programs that trade in microseconds
 - ▶ Budish, Cramton & Shim (2015) argue that continuous trading creates a speed arms race and propose frequent batch auctions instead
 - ▶ algorithmic trading and its effect on liquidity: Chapter 10 (Market Microstructure)
- Spreads, impact and fragmentation add noise to observed prices, most visible at high frequency (Chapter 9)

Recap: Market Structure

- Trading happens on exchanges, OTC with dealers, in dark pools and, for crypto, on centralised and decentralised venues
- The limit order book stores waiting orders; market orders consume them
- The spread and the depth measure how costly it is to trade
- Auctions set the reference prices; continuous trading rewards speed

From the Trading Floor to Screens



NYSE trading floor, 2011

Photo: Carol M. Highsmith (2011);
public domain; Wikimedia Commons



Electronic displays, Chicago Board of Trade, 1992

Photo: Lars Plougmann (1992); CC BY-SA 2.0;
Wikimedia Commons

- **Open outcry:** traders met on the floor and matched orders by voice and hand signals
- Electronic limit order books gradually replaced most floor trading
- Today most volume is matched by computers (market microstructure, Chapter 10)

Crashes Before Our Data: Black Tuesday, 1929

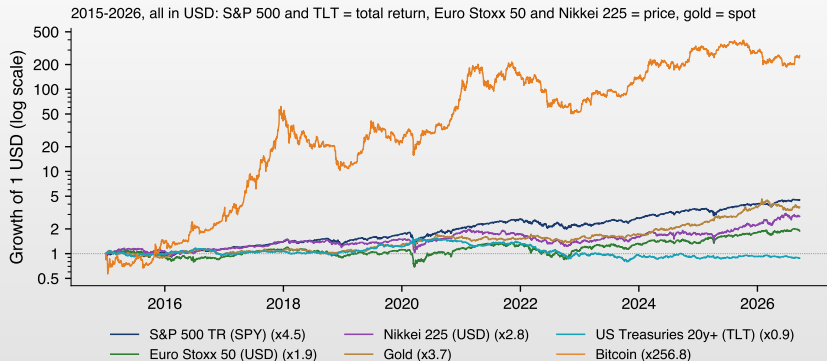


Crowd outside the NYSE, 29 October 1929

Photo: U.S. government (1929); public domain;
[Wikimedia Commons](#)

- 29 October 1929 (**Black Tuesday**): crowds gather outside the New York Stock Exchange during the crash
- The crash of October 1929 opened the Great Depression of 1929–1933
- Long, deep drawdowns are not new: the next slides measure the 2000–2026 episodes with the same tools
- Bubbles and crashes are modelled in Chapter 17

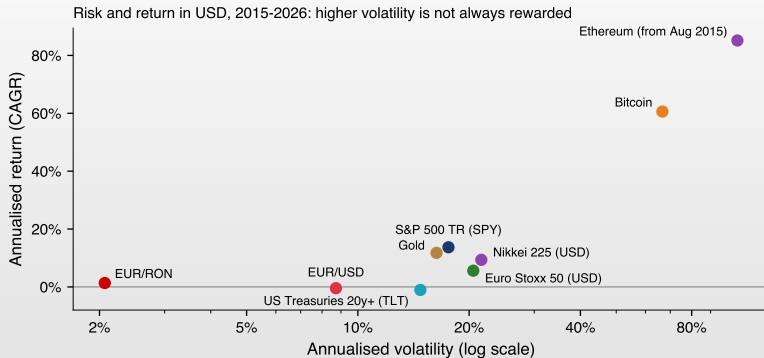
Asset Classes since 2015



Reading the Cross-Asset Chart

- Each line shows the value of 1 USD invested on 2 Jan 2015; every series is in USD, on a log scale
 - ▶ Euro Stoxx 50 and Nikkei 225 converted with daily EUR/USD and USD/JPY rates from [FRED](#)
- S&P 500 with dividends reinvested (SPY, total return) x4.5; gold x3.7; Nikkei 225 x2.9; Euro Stoxx 50 x1.9
- **TLT** (iShares 20+ Year Treasury Bond ETF, long US government bonds, total return) ended **below** its 2015 level: x0.9
- Bitcoin multiplied by about 257, from 315 USD to 80 901 USD
- Euro Stoxx 50 and Nikkei 225 are price indices: their dividends are missing
 - ▶ for the S&P 500 the gap is 1.8 pp a year: 13.7% with dividends (SPY) vs 11.9% for the price index

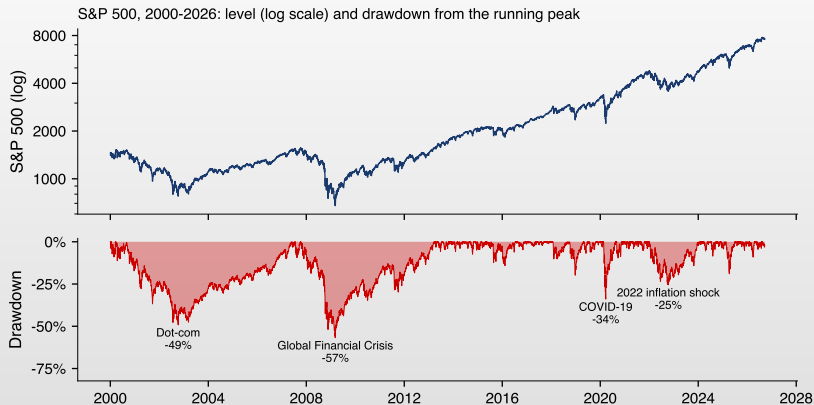
Risk and Return, 2015–2026



Reading the Risk–Return Chart

- Vertical axis: CAGR; horizontal axis: annualised volatility of daily log returns (both defined in Section 2)
- All in USD. S&P 500 with dividends (SPY): 13.7% a year, 17.6% volatility; Nikkei 225: 9.5%, 21.6%; Euro Stoxx 50: 5.7%, 20.5%
- Gold: 11.8% a year with 16.3% volatility, slightly below the S&P 500 (weekday quotes only; weekend prints removed)
- Bitcoin: 60.6% a year, 66.7% volatility; Ethereum (from Aug 2015): 85.2% and 106%
- EUR/RON: volatility 2.1% a year, the profile of a managed currency
- Ex post only: returns earned in the past are not forecasts

Crises and Drawdowns

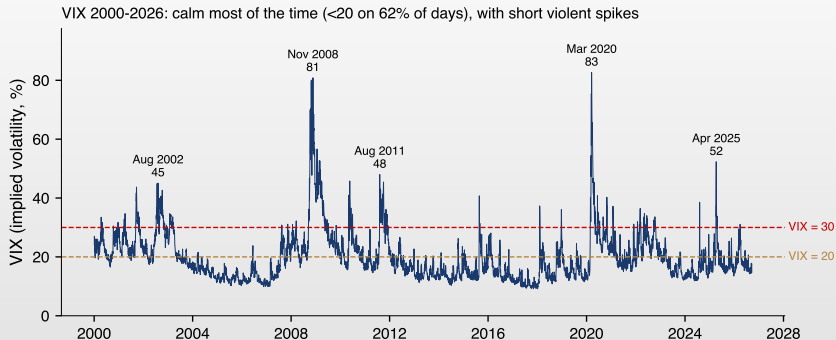


Reading the Drawdown Chart

- Lower panel: the drawdown $DD_t = P_t / \max_{s \leq t} P_s - 1$ of the S&P 500
- Four large troughs since 2000
 - ▶ −49% (Oct 2002): the end of the dot-com bubble
 - ▶ −57% (Mar 2009): the global financial crisis
 - ▶ −34% (Mar 2020): the COVID-19 crash, the fastest of the four
 - ▶ −25% (Oct 2022): the inflation and interest-rate shock
- Recovery can take years: after 2009 the index needed until March 2013 to regain its 2007 peak

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Volatility Regimes: the VIX



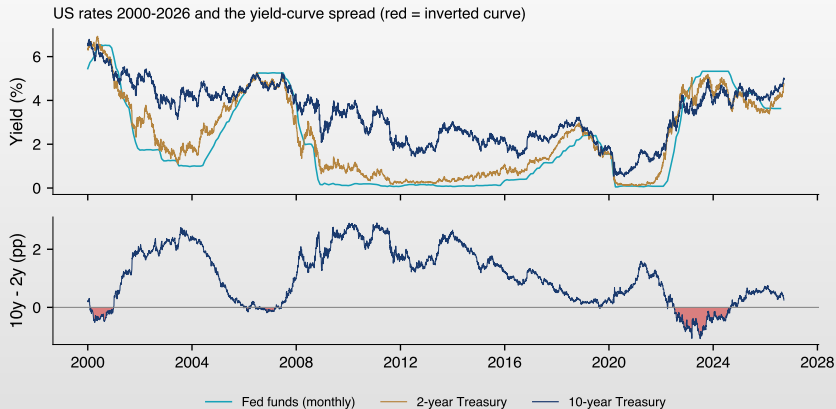
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Reading the VIX Chart

- **VIX**: the CBOE Volatility Index, computed by the Chicago Board Options Exchange
 - ▶ the volatility of the S&P 500 over the next 30 days that is **implied** by option prices, in % per year
 - ▶ often called the “fear index”
- Since 2000 the VIX was below 20 on 62% of days and above 30 on 9%
- Largest peaks: 82.7 (16 Mar 2020), 80.9 (20 Nov 2008), 52.3 (8 Apr 2025)
- Calm periods are long, stress periods short and intense: volatility comes in regimes (Chapters 5 and 9)

 MFM_ch0_drawdowns_vix

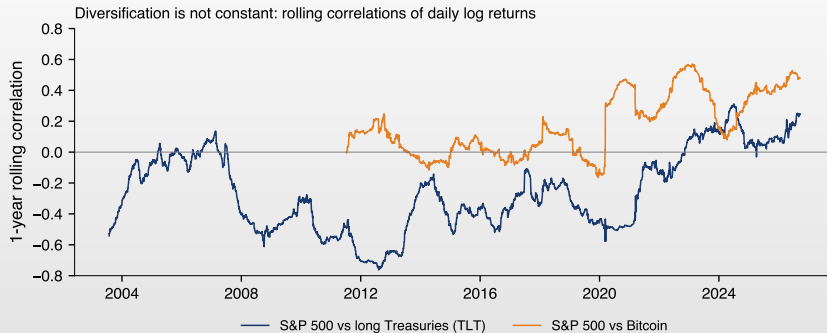
Interest Rates and the Yield Curve



Reading the Rates Chart

- Data: [FRED](#) (Federal Reserve Economic Data, St. Louis Fed)
 - ▶ 10-year and 2-year US Treasury yields; the **fed funds rate**, the overnight rate targeted by the US central bank
- The 10y – 2y spread first turned negative briefly on 1–4 Apr 2022 (2 trading days)
 - ▶ persistent inversion from 6 Jul 2022 to 26 Aug 2024: 537 trading days, minimum -1.08 pp (3 Jul 2023)
- 17 Sep 2026: 10-year 4.94%, 2-year 4.67%, spread $+0.27$ pp = 27 bp; fed funds 3.63% (Aug 2026)
- Rates set the discount factor of every asset in this course

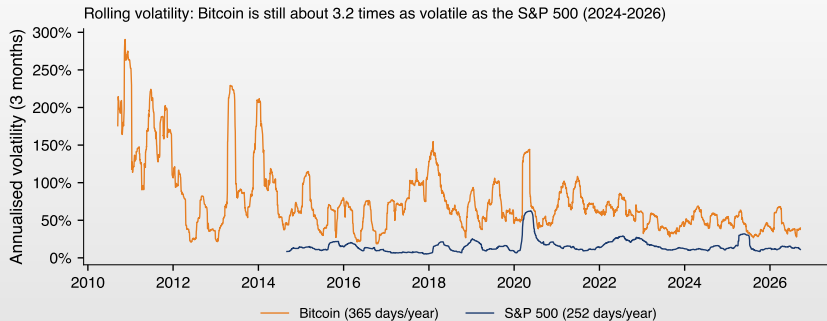
Diversification Is Not Constant



Reading the Correlation Chart

- 1-year (252-day) rolling correlations of daily log returns; prices joined on common trading days first, then returns
- Stocks vs. long Treasuries (S&P 500 vs. TLT)
 - ▶ mean of the 252-day rolling correlation: -0.43 in 2010–2020, $+0.08$ in 2022–2026 (whole-period correlation: -0.47 , $+0.11$; Seminar 0)
 - ▶ after the 2022 inflation shock bonds hedged equities much less
- Stocks vs. Bitcoin
 - ▶ mean of the 252-day rolling correlation: 0.01 in 2016–2019, 0.38 in 2022–2026 (whole-period 2022–2026: 0.42)
 - ▶ crypto now moves more with equities than it did before 2020
- Portfolio models that assume a fixed correlation miss these shifts (Chapters 4 and 6)

Volatility Differs by an Order of Magnitude



Reading the Rolling Volatility Chart

- Rolling 3-month (63 trading days) annualised volatility
 - ▶ Bitcoin annualised with $\sqrt{365}$, the S&P 500 with $\sqrt{252}$
- Bitcoin: average 65% in 2015–2017, 46% in 2024–2026
- S&P 500 in 2024–2026: 14%, so Bitcoin is still about **3.2 times** as volatile
- Volatility changes over time for both: a constant-volatility model is a poor description

 MFM_ch0_correlations

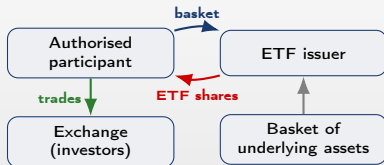
Recap: the 2000–2026 Landscape

- ▣ Returns differ widely across assets and are not a reward for volatility alone
- ▣ Deep drawdowns recur, and recoveries can take years
- ▣ Volatility comes in regimes: long calm periods, short stress periods
- ▣ Correlations change with the macro regime, so diversification is not guaranteed
- ▣ Crypto remains several times more volatile than equities, and more correlated with them than before

Exchange-Traded Funds

- ▣ An **ETF** is an investment fund whose shares trade on an exchange like a stock
- ▣ **Net asset value (NAV)**: the value of the fund's holdings per share, computed once a day
- ▣ Most ETFs are **passive**: they replicate an index at low cost
- ▣ Advantages: low fees, trading during the day, diversification in one security
 - ▶ in the US also tax efficiency: in-kind redemptions let the fund avoid distributing capital gains (a US tax rule, not a general feature)
- ▣ ETFs now package equities, bonds, commodities and, in the US since January 2024, spot Bitcoin

How ETF Creation and Redemption Work



- **Authorised participant (AP):** a large dealer allowed to deal directly with the issuer
- **Creation:** the AP delivers the basket of underlying assets and receives new ETF shares
- **Redemption:** the AP returns ETF shares and receives the basket
- This two-way channel keeps the market price close to the NAV

Worked Example: Arbitrage Keeps the ETF Price Near NAV

- Illustrative numbers: $NAV = 100$, the ETF trades at 100.50
- **Premium:** $(P - NAV)/NAV = 0.50/100 = 0.5\%$
- The AP
 - ▶ buys the basket for 100 and delivers it to the issuer
 - ▶ receives one new ETF share and sells it on the exchange for 100.50
 - ▶ earns 0.50 per share before costs
- The extra ETF shares push the price down until the premium disappears
- At a **discount** ($P < NAV$) the AP does the reverse: buys ETF shares and redeems them

ETFs and Volatility

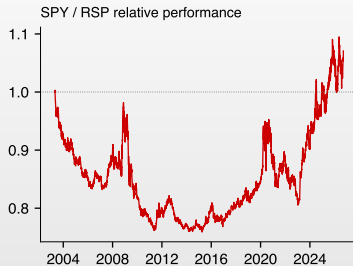
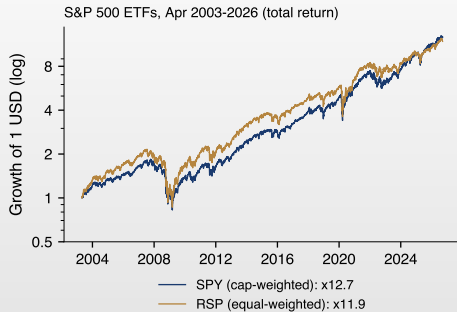
- ETF shares are traded more often and by faster investors than the underlying stocks
- Arbitrage transmits ETF price shocks to the underlying stocks
- Ben-David, Franzoni & Moussawi (2018): stocks with higher ETF ownership are more volatile
- For modelling: the growth of passive investing can change the statistical behaviour of the stocks it holds

Cap-Weighted versus Equal-Weighted Indices

Definition 7 (Index weights)

- Capitalisation weights: $w_i = \frac{MC_i}{\sum_{j=1}^n MC_j}$, with market capitalisation $MC_i = \text{price} \times \text{number of shares}$
 - Equal weights: $w_i = 1/n$ for every stock, rebalanced periodically
-
- SPY** (SPDR S&P 500 ETF): the S&P 500 with capitalisation weights
 - RSP** (Invesco S&P 500 Equal Weight ETF): the same 500 stocks, each with weight $1/500$
 - Ratio SPY/RSP rising: the largest companies are outperforming the average company

Market Concentration: SPY versus RSP



Reading the Concentration Chart

- ▣ Both ETFs are total return (adjusted close), since RSP's launch in April 2003
- ▣ Over the whole period the two grew similarly: SPY $\times 12.7$, RSP $\times 11.9$
- ▣ The SPY/RSP ratio rose by 31% since January 2023, above its 2003 starting level
- ▣ Interpretation: a handful of very large companies drove the index in 2023–2026

 MFM_ch0_concentration

Why Concentration Matters for Modelling

For investors

- ▣ A cap-weighted index is less diversified than its number of stocks suggests
- ▣ Index risk depends on a few firms and one theme
- ▣ Passive flows buy the largest firms mechanically

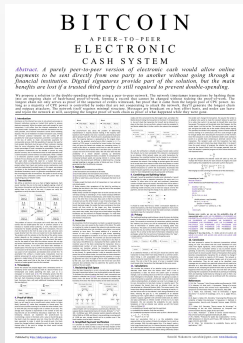
For statisticians

- ▣ The effective number of independent bets is small (Chapter 4)
- ▣ Factor exposures shift over time (Chapter 3)
- ▣ Bubbles in a few names can dominate the index (Chapter 17)

Recap: Passive Investing

- ETFs give cheap, tradable access to whole indices and asset classes
- Creation and redemption by authorised participants keep ETF prices close to NAV
- Passive investing grows with capitalisation weights, which concentrates the index in the largest firms
- Concentration reduces diversification and changes the risk of the index

Where It Started: the Bitcoin White Paper



First page of the Bitcoin white paper (2008)

Satoshi Nakamoto (2008); CC0; Wikimedia Commons

- 31 October 2008: *Bitcoin: A Peer-to-Peer Electronic Cash System*, by the pseudonymous (real identity unknown) Satoshi Nakamoto [Nakamoto \(2008\)](#)
- A public ledger (the **blockchain**) kept by the network itself, without a trusted intermediary
- **Proof of work**: participants spend computing power to add blocks, which makes rewriting the history costly
- From this paper to spot ETFs, stablecoins and tokenisation: the next slides

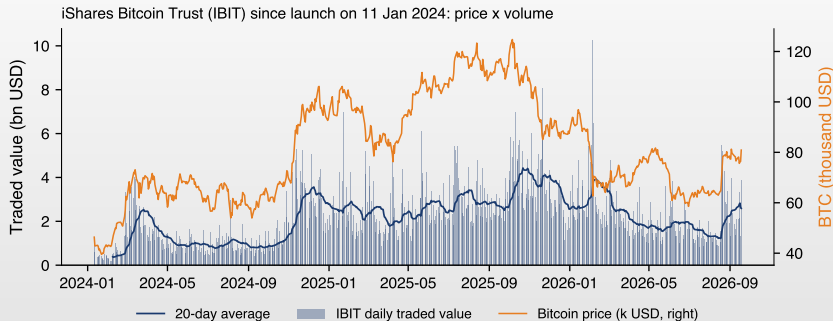
Cryptocurrencies: Basic Terms

- **Blockchain:** a shared ledger of transactions, copied on many computers and extended in blocks
- **Bitcoin** (2009): the first cryptocurrency, with a fixed supply rule (at most 21 million coins)
- **Ethereum:** a blockchain that runs programs, called **smart contracts**; its coin is Ether
- **Token:** a digital unit recorded on a blockchain, representing a coin, a claim or a right
- Trading is continuous, global and fragmented across many exchanges

Cryptocurrencies as an Asset Class

- ▣ Härdle, Harvey & Reule (2020): statistical properties, pricing and regulation of cryptocurrencies
- ▣ Pele et al. (2023): tail, memory and moment factors of returns set cryptos apart from stocks, FX, commodities, bonds and real estate
- ▣ Makarov & Schoar (2020): large and persistent price differences across exchanges and countries
- ▣ Our data: correlation with equities (mean of the 252-day rolling correlation) rose from about 0 (2016–2019) to 0.38 (2022–2026)
- ▣ Chapter 16 covers crypto indices (CRIX, the CRYPTO Index), stablecoins, AMMs and on-chain data in depth

Spot Bitcoin ETFs



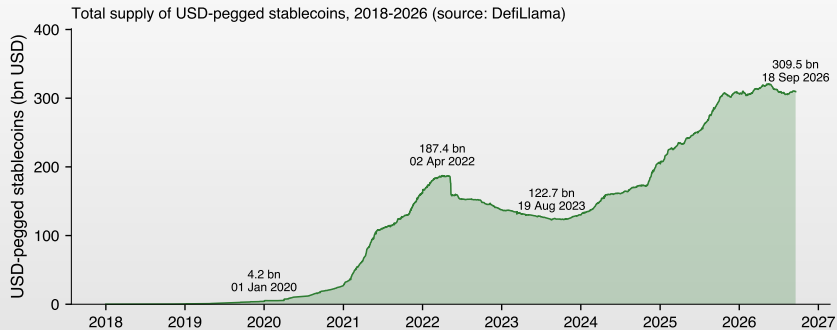
Reading the Spot Bitcoin ETF Chart

- **Spot** ETF: holds the asset itself (bitcoins), not futures contracts on it
- **ETP**: exchange-traded product, the general category that includes ETFs
- **SEC (2024)**: approval of spot Bitcoin ETPs in the United States, 10 January 2024
- **IBIT** (iShares Bitcoin Trust), trading since 11 Jan 2024
 - ▶ daily traded value = price $\times$ number of shares traded
 - ▶ median 1.97 bn USD per day, peak 10.3 bn USD (5 Feb 2026)

What Is a Stablecoin Peg?

- ▣ **Stablecoin:** a token designed to keep a fixed value, usually 1 USD; usually, but not always, backed by reserves
- ▣ **Peg:** the target exchange rate, e.g. 1 token = 1 USD
 - ▶ **reserve-backed:** the issuer holds cash and short-term government bonds and redeems tokens at 1 USD
 - ▶ **algorithmic:** the peg relies on a second token and trading rules, without full reserves
- ▣ **Depeg:** the market price moves away from the peg
- ▣ Our data, Tether (USDT), 2019–2026: within $\pm 0.5\%$ of 1 USD on 93% of days and within $\pm 1\%$ on 98%
 - ▶ extremes 0.974 and 1.054 USD, both in the COVID-19 panic of March 2020
- ▣ The algorithmic stablecoin TerraUSD lost its peg and collapsed in May 2022

Stablecoin Supply



Reading the Stablecoin Chart

- **Supply:** the total value of USD-pegged stablecoins in circulation, from [DefiLlama](#), a public DeFi data platform
- 4.2 bn USD (1 Jan 2020), peak 187.4 bn (2 Apr 2022), trough 122.7 bn (19 Aug 2023), 309.5 bn (18 Sep 2026)
- The 2022–2023 drop followed the collapse of TerraUSD and the crypto crash of 2022
- Regulated in the European Union by [MiCA, Regulation \(EU\) 2023/1114](#) (Markets in Crypto-Assets) and in the US by the [GENIUS Act, Public Law 119-27](#)

Tokenisation of Real-World Assets

Idea

- ▣ **RWA**: real-world assets, such as bonds, fund shares, deposits or commodities
- ▣ A token on a shared ledger represents a claim on such an asset
- ▣ Settlement and transfer become programmable and near-instant
- ▣ Examples: tokenised money-market funds and Treasury funds

- ▣ Statistical question: do tokenised assets inherit the properties of the underlying, or those of crypto markets?

Central-bank view

- ▣ **BIS**: Bank for International Settlements, the bank of central banks
- ▣ **BIS (2023)**: a “unified ledger” combining central bank money, deposits and assets
- ▣ **BIS (2025)**: stablecoins fall short of the three tests for money
 - ▶ singleness (accepted at par by all), elasticity (supply follows payment needs), integrity (resists financial crime)

Regulation of Digital Assets

	European Union	United States
Framework	MiCA, Regulation (EU) 2023/1114	GENIUS Act, Public Law 119-27 for payment stablecoins
Crypto ETPs	exchange-traded notes on crypto	spot Bitcoin ETPs approved in January 2024 (SEC (2024))
Supervisors	ESMA, EBA, national authorities	SEC, CFTC, banking regulators

- ESMA: European Securities and Markets Authority; EBA: European Banking Authority
- SEC: Securities and Exchange Commission; CFTC: Commodity Futures Trading Commission
- For modelling: regulatory dates are natural break points and event-study dates

Recap: Digital Assets

- Cryptocurrencies are a statistically distinct asset class, mainly through their heavy tails
- Spot ETFs and stablecoins connected crypto to mainstream finance
- Stablecoins hold their peg most of the time, but pegs can break
- Tokenisation brings traditional assets onto blockchains; regulation (MiCA, GENIUS Act) shapes who can hold them

The Bucharest Stock Exchange, 1882–1948

- 1 Dec 1882: the securities and commodities exchange opens under a royal decree of King Carol I [BVB \(history\)](#)
- Early shocks [BVB press release \(2017\)](#)
 - the 1907 peasants' revolt and the Balkan war of 1912–1913
 - closed during the First World War
- After reopening: about 7 years of strong growth, then about 7 years of accelerated decline, overlapping the Great Depression of 1929–1933 [BVB press release \(2017\)](#)
- 1929: a new law on exchanges, with the explanatory memorandum of minister Virgil Madgearu [Lege asupra burselor \(1929\)](#)
- 1948: trading stops with the communist regime and the nationalisation of the economy [BVB \(history\)](#)



Share of UDR (Uzinele și Domeniile Reșița), June 1926. Scan: [EDHAC e.v.](#); public domain; [Wikimedia Commons](#)

The Palatul Bursei: Home of the Exchange



Photo: Nicolae Ionescu (1928); public domain; Wikimedia Commons



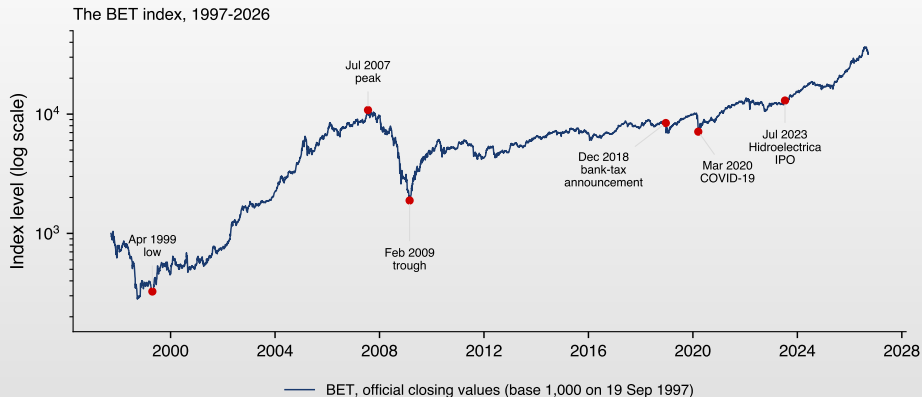
Photo: Chainwit. (2023); CC BY 4.0; Wikimedia Commons

- Built 1906–1912, architect Ștefan Burcuș, Str. Ion Ghica 4 [Palatul Bursei](#)
- Seat of the Exchange and of the Bucharest Chamber of Commerce and Industry
- From 1955: the State Central Library (today the National Library)
- Left: stockbrokers at the entrance, March 1928; right: the building in 2023

Interruption and Rebirth, 1948–2023

- ▣ **1948–1995**: almost 50 years without a stock exchange [BVB press release \(2017\)](#)
- ▣ **20 Nov 1995**: first session of the re-established exchange: 905 shares of 6 listed companies [BVB press release \(2017\)](#)
- ▣ **1997**: launch of **BET** (Bucharest Exchange Trading index) on 19 Sep 1997 at 1 000 points, with 10 companies; the basket may hold 10–20 and today has 20 [BVB \(indices\)](#)
 - ▶ **BET-FI**: the first sector index, for the financial investment companies (SIFs, listed in 1999)
- ▣ **2010**: BVB lists its own shares on its **regulated market** (an EU-authorized trading venue) [BVB \(history\)](#)
- ▣ **Sep 2020**: **FTSE Russell**, a global index provider, reclassifies Romania from Frontier to (Secondary) Emerging [BVB \(history\)](#)
- ▣ **Jul 2023**: Hidroelectrica lists after the largest **IPO** (initial public offering, a first sale of shares to the public) at BVB, EUR 1.9 bn [BVB \(history\)](#)

The BET Index, 1997–2026



Reading the BET Chart

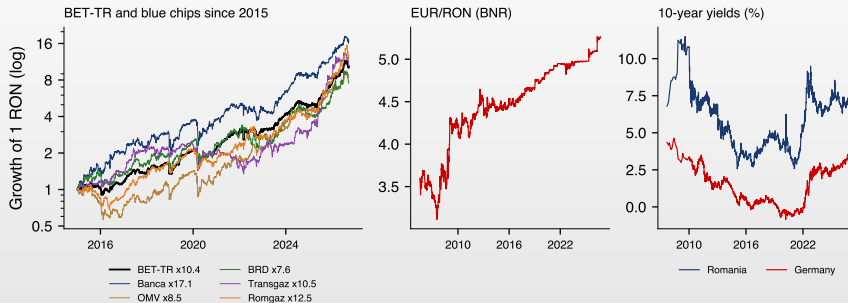
- One series: official BET closing values, 19 Sep 1997 – 18 Sep 2026; log scale, so equal distances are equal percentage changes
- Launch 1000, low 325.78 (21 Apr 1999), peak 10 813.59 (24 Jul 2007), trough 1 887.14 (25 Feb 2009): a drawdown of -82.5%
 - ▶ the 2007 peak was regained only on 16 Mar 2021, after 13.6 years
- 19 Dec 2018: -11.2% in one day (8 419.19 to 7 475.22), the day after the bank-tax announcement (Chapter 1)
- Mar 2020 (COVID-19): -31.1% from the February high to 23 Mar 2020
- 18 Sep 2026: 32 757.81, $\times 32.8$ since launch, a CAGR of 12.8% over 29 years (record 36 487.21 on 19 Aug 2026)
- Largest daily move: 7 Jan 2009, -12.3% (-13.1% log), after $+4.8\%$ the day before (official BVB daily archive BVB (indices))

 MFM_ch0_bvb_history

The Bucharest Stock Exchange

- **BVB**: Bursa de Valori București, the Bucharest Stock Exchange
- **BET**: the index of the most traded Romanian companies (blue chips); **BET-TR**: its total-return version, with reinvested dividends
- Largest companies: banks (Banca Transilvania, BRD), energy (OMV Petrom, Romgaz, Hidroelectrica), utilities (Transgaz)
- Classified as **Secondary Emerging** by [FTSE Russell \(2026\)](#), the index provider
- **TVBETETF**: an ETF traded on BVB that replicates BET, the whole index in one security

Romanian Equities, the Leu and Government Yields



Reading the Romanian Chart

- ▣ Our data: BET-TR from Sep 2014, Hidroelectrica from its listing in Jul 2023
- ▣ Growth since Jan 2015, dividend-adjusted
 - ▶ BET-TR x10.4, Banca Transilvania x17.1, Romgaz x12.5, Transgaz x10.5, OMV Petrom x8.5, BRD x7.6
- ▣ EUR/RON, official reference rate of the [BNR](#) (Banca Națională a României, the National Bank of Romania)
 - ▶ 3.60 (4 Jul 2005) to 5.26 (Sep 2026), volatility 2.1% a year since 2015
 - ▶ the new leu (RON) replaced the old leu on 1 Jul 2005 (1 RON = 10 000 ROL); the BNR archive states earlier rates in RON
- ▣ 10-year government yields, 18 Sep 2026: Romania 7.33%, Germany 3.53%
 - ▶ spread of 3.8 pp, the extra yield investors require for Romanian risk; peak 8.3 pp in Mar 2009

Data Used in This Course

Data	Content	Access in Python
Daily market data (data/market)	indices, stocks, ETFs, FX, crypto, yields	<code>pandas.read_csv</code>
FRED	US rates, macro series	public CSV
BNR	official RON reference exchange rates	public XML
DefiLlama	stablecoins, DeFi	public JSON

- Market files: one CSV per symbol, columns date, open, high, low, close, adjusted_close, volume
- All loading goes through one function, `fetch_daily(symbol, source)`
- Rule: `adjusted_close` for stocks and ETFs, `close` for indices, FX, crypto and yields
- EUR/RON always from the BNR reference rate; the market file serves only the data-pitfall example

A Minimal Python Workflow

```
import numpy as np, pandas as pd

URL = ('https://raw.githubusercontent.com/'
      'danpele/MFM/main/data/market/')

def load(sym, col='close'):
    d = pd.read_csv(URL + sym + '.csv',
                    index_col='date', parse_dates=True)
    return d[col].rename(sym)

spx = load('GSPC.INDX')          # index: close
spy = load('SPY.US', 'adjusted_close') # ETF
btc = load('BTC-USD.CC')         # 24/7
r_spx = np.log(spx).diff()       # one series
# joint analysis: join PRICES on common
# days first, then take returns
p = pd.concat([spx, btc], axis=1).dropna()
r = np.log(p).diff().dropna()
dd = spx / spx.cummax() - 1      # drawdown
```

- One series: returns on its own calendar, $\text{np.log}(s).\text{diff}()$ (the r_t of Section 2)
- Joint analysis (correlations, portfolios): join the *prices* on common days, then take returns, so both cover the same interval (Friday → Monday)
- Returns taken before the join would pair a Bitcoin Sunday → Monday move with an S&P 500 Friday → Monday move and lose Bitcoin's weekends
- $\text{spx} / \text{spx.cummax}() - 1$: the drawdown DD_t
- Crypto trades every day: annualise with 365, not 252

Data Pitfalls: Real Examples

- ▣ **Bad tick:** a wrong price in a data file, typically a spike that is reversed the next day
- ▣ EUR/RON in the market file (data/market), used only for this example
 - ▶ daily log changes of $\pm 15\%$ on 13–14 Aug 2025 and $\pm 12\%$ in early Jan 2022, each reversed the next day
 - ▶ annualised volatility 2015–2026: **12.1%** with these ticks, **2.1%** with the BNR reference rates
- ▣ Romania 10-year yield: a 0.00% print on 24 Dec 2013 and stale values on Sundays in late 2022
- ▣ A handful of bad ticks inflated the volatility estimate more than five-fold: cleaning data is part of the model

Data Checklist

- Plot the raw series and list the largest returns
- Compare two sources when possible
- Check adjustments: dividends, splits, redenominations
- Drop non-trading days and isolated outliers, and document every change
- Compute returns on each asset's own calendar before aligning assets
- Watch survivorship: delisted firms vanish from today's index lists, so past samples look too good

Key Takeaways

- Markets in 2026 combine exchanges, OTC dealers, ETFs, algorithms and blockchains
- Across 2000–2026, the same patterns recur:
 - ▶ deep drawdowns and short volatility spikes
 - ▶ correlations that change with the macro regime
 - ▶ very different volatility levels across assets
- Passive investing increased concentration; crypto moved closer to mainstream markets via ETFs and stablecoins
- Data quality is a modelling decision: sources, adjustments and calendars change the results

Next: Chapter 1, Stylised Facts

Questions for Chapter 1

- Why do we work with returns instead of prices?
- Are daily returns normally distributed? What about monthly returns?
- Why do large price changes cluster in time?
- Do these facts hold for the S&P 500, Bitcoin, BVB stocks and EUR/RON alike?

Further Study on Quantinar

Advanced courses on quantinar.com

-  Digital Transformation on Finance
-  Cryptos, NFTs, Digital Assets
-  Introduction to Blockchain and Cryptocurrencies
-  Statistics of Financial Markets

References (1/2)

- ▣ Ben-David, I., Franzoni, F., Moussawi, R. (2018). Do ETFs increase volatility? *Journal of Finance*, 73(6), 2471–2535.
- ▣ Brandimarte, P. (2018). *An Introduction to Financial Markets: A Quantitative Approach*. Wiley.
- ▣ Budish, E., Cramton, P., Shim, J. (2015). The high-frequency trading arms race: frequent batch auctions as a market design response. *Quarterly Journal of Economics*, 130(4), 1547–1621.
- ▣ Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*. Princeton University Press.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction* (5th ed.). Springer.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). *Applied Quantitative Finance* (3rd ed.). Springer.
- ▣ Härdle, W.K., Harvey, C.R., Reule, R.C.G. (2020). Understanding cryptocurrencies. *Journal of Financial Econometrics*, 18(2), 181–208.
- ▣ Härdle, W.K., Simar, L. (2019). *Applied Multivariate Statistical Analysis*. Springer.

References (2/2)

- ▣ Makarov, I., Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2), 293–319.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ BVB (2017). 22 de ani de la prima tranzacție la Bursa de Valori București. Press release, 20 November 2017.
- ▣ BVB. BVB indices: BET, launched 19 September 1997. Bucharest Stock Exchange.
- ▣ BIS (2023). Blueprint for the future monetary system: improving the old, enabling the new. Annual Economic Report, Chapter III.
- ▣ BIS (2025). The next-generation monetary and financial system. Annual Economic Report, Chapter III.
- ▣ ESMA. Markets in Crypto-Assets Regulation (MiCA), Regulation (EU) 2023/1114.
- ▣ FTSE Russell (2026). FTSE Equity Country Classification, Interim Announcement, April 2026.
- ▣ SEC (2024). Statement on the Approval of Spot Bitcoin Exchange-Traded Products, 10 January 2024.
- ▣ U.S. Congress. Guiding and Establishing National Innovation for U.S. Stablecoins Act (GENIUS Act), Public Law 119-27.
- ▣ Code and data: github.com/danpele/MFM/tree/main/Quantlets/Ch_00
- ▣ Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system. White paper.